US Natural Disaster Temporal Analysis & Visualization

Milestone 4: Incident Analysis & Federal Assistance Correlation

Uma Maheswar Reddy V

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1. Project description

The **US Natural Disaster Temporal Analysis** project is a data-driven study designed to analyze how disaster declarations evolve over time across the United States. The primary goal is to identify patterns, trends, and seasonality in the data to understand the frequency and nature of these events.

Objectives: The goal of this milestone is to move beyond "where" and "when" disasters occur to analyze the specific nature of the incidents and the resulting federal financial response.

- **Incident Characterization:** Quantify the distribution of over 20 disaster types to identify the primary drivers of federal declarations.
- **Geospatial Correlation:** Map specific disaster types to individual states to identify regional "Disaster Signatures" (e.g., Fire in the West vs. Hurricanes in the South).
- **Resource Allocation Analysis:** Evaluate the relationship between incident types and the activation of Individual Housing (IH) and Public Assistance (PA) programs.

2. DataSet Preparation

To ensure the accuracy of the incident analysis, the following data processing steps were performed:

- **Column Renaming:** Standardized individualsAndHouseholdsProgram to ihProgramDeclared and publicAssistanceProgram to paProgramDeclared for improved readability.
- **Data Type Validation:** Enforced integer types on assistance columns to allow for mathematical summation and aggregation.
- **Feature Extraction:** Ensured that year and month features from Milestone 2 and the state codes from Milestone 3 were fully integrated.
- **Assistance Mapping:** Since assistance columns contained string values ("Yes"/"No"), a mapping dictionary was applied (Yes=1, No=0).

3. Environment Setup

To conduct this cumulative analysis, a robust data science environment was configured using Python and several specialized libraries:

- **Pandas:** Utilized for high-performance data manipulation, including pivot tables and multi-index grouping.
- **Plotly Express:** Employed for generating interactive, multi-dimensional visualizations such as Treemaps and Grouped Bar Charts.
- **Seaborn & Matplotlib:** Used for advanced statistical plotting, specifically the State-Incident Heatmap to visualize declaration density.
- **VS Code / Jupyter:** The development environment used to integrate code, visualizations, and markdown documentation.

Following are the requirements we have installed for the complete project but for this specific milestone we have used only Pandas, Matplotlib and Jupyter Notebook.

```
numpy
pandas
matplotlib
seaborn
plotly
geopandas
folium
jupyter
```

4. Incident Type Distribution Analysis

This section quantifies the frequency and composition of natural disasters across the entire 1953–2017 timeline.

historical frequency of federal disaster declarations and understand the categorical composition of emergency events.

4.1 Quantitative Overview

The distribution analysis covers 64 years of federal records, totaling nearly 50,000 declarations. By aggregating data by `incidentType`, we move from a general timeline to a specific understanding of which environmental threats demand the most federal attention.

4.2 Categorical Dominance

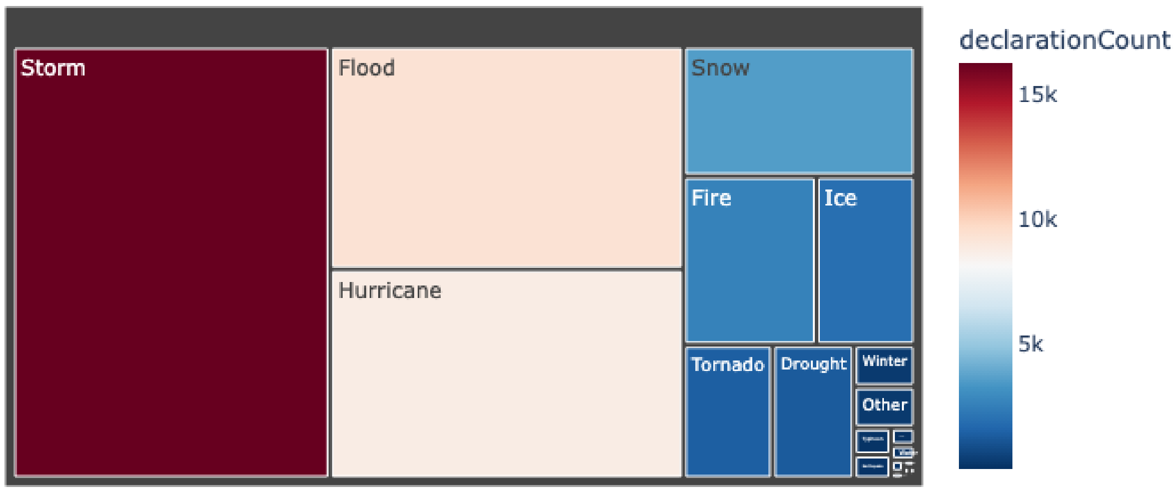
- **Atmospheric Events:** "Storms" (16,250) represent the highest volume of data. This category is broad, encompassing severe thunderstorms, coastal gales, and wind events that occur year-round across the entire continental United States.
- **Hydrological Threats:** "Floods" (9,317) are the second most frequent. Unlike storms, floods often require longer-term federal recovery cycles and represent a massive portion of the "Public Assistance" budget.
- **Seasonal Extremes:** "Hurricanes" (8,764) and "Snow" (3,565) show significant seasonal spikes. While hurricanes have fewer total declarations than storms, they often result in more concentrated, high-impact county-level declarations within a single week.

4.3 Composition Insights

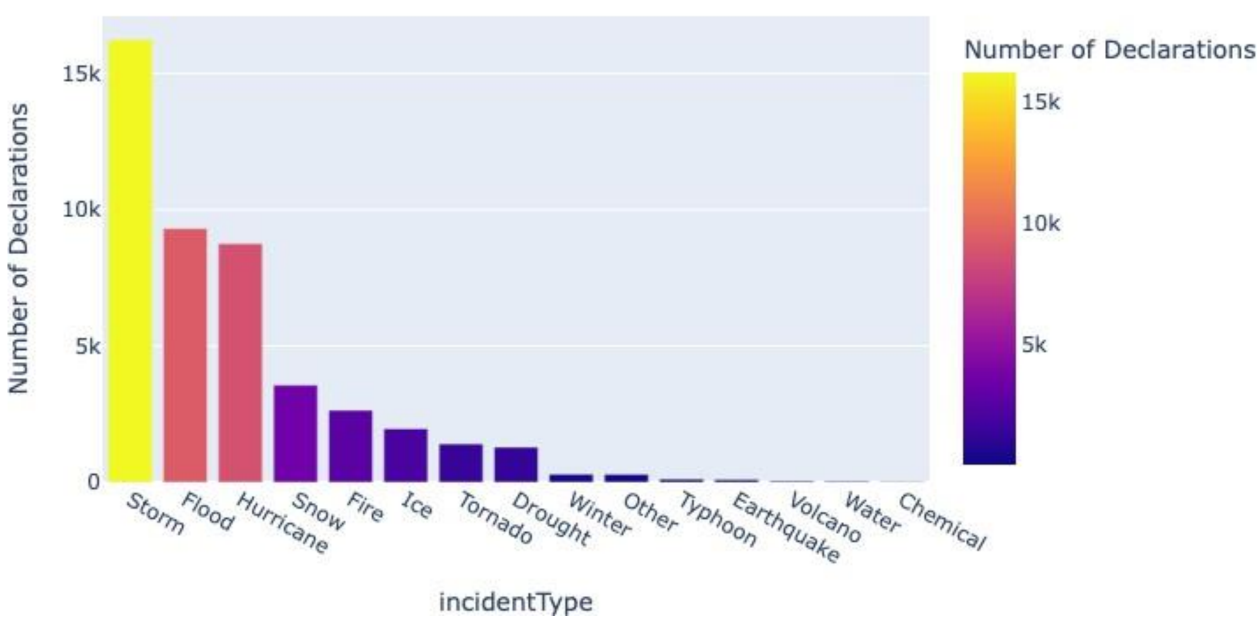
Using the **Treemap** visualization, we observe that the "Top 3" incidents (Storms, Floods, Hurricanes) make up nearly 75% of the entire dataset. This indicates that the Federal Emergency Management Agency (FEMA) is primarily a weather-response organization, with geological events like Earthquakes and Volcanoes acting as rare, low-frequency outliers.

Section 4.4 (Incident Distribution):

Milestone 4: US Natural Disaster Composition (Treemap)



Top 15 Most Frequent Disaster Types



5. State vs Incident Type Analysis

Goal: To identify regional "Disaster Signatures" by correlating incident types with specific geographic locations.

5.1 Geographic Correlation Logic

Disasters are not distributed uniformly across the U.S.; they follow specific climactic and geological belts. By creating a pivot table of `state` vs. `incidentType`, we can mathematically prove which states are "magnets" for specific catastrophes.

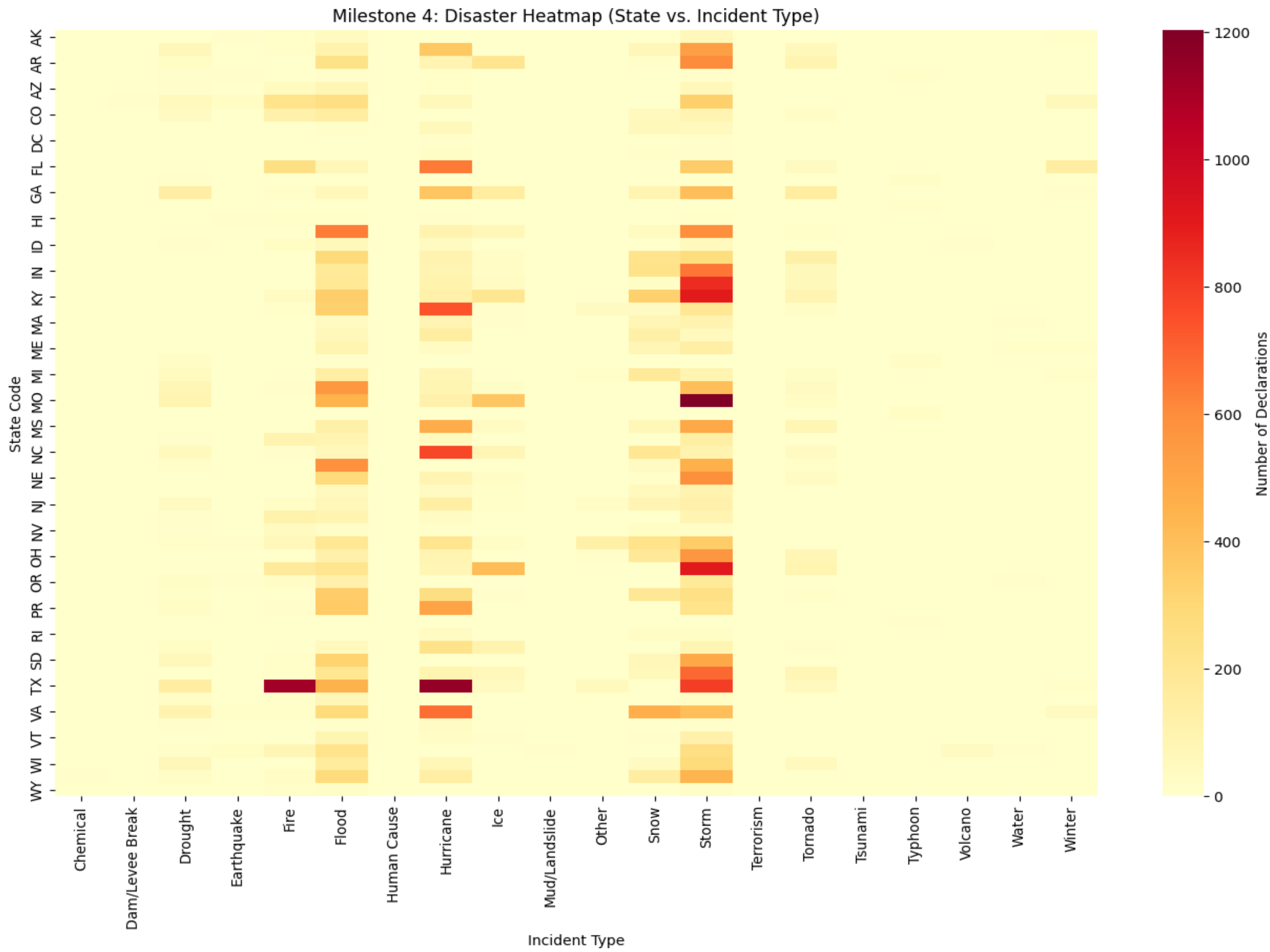
5.2 Regional "Signatures"

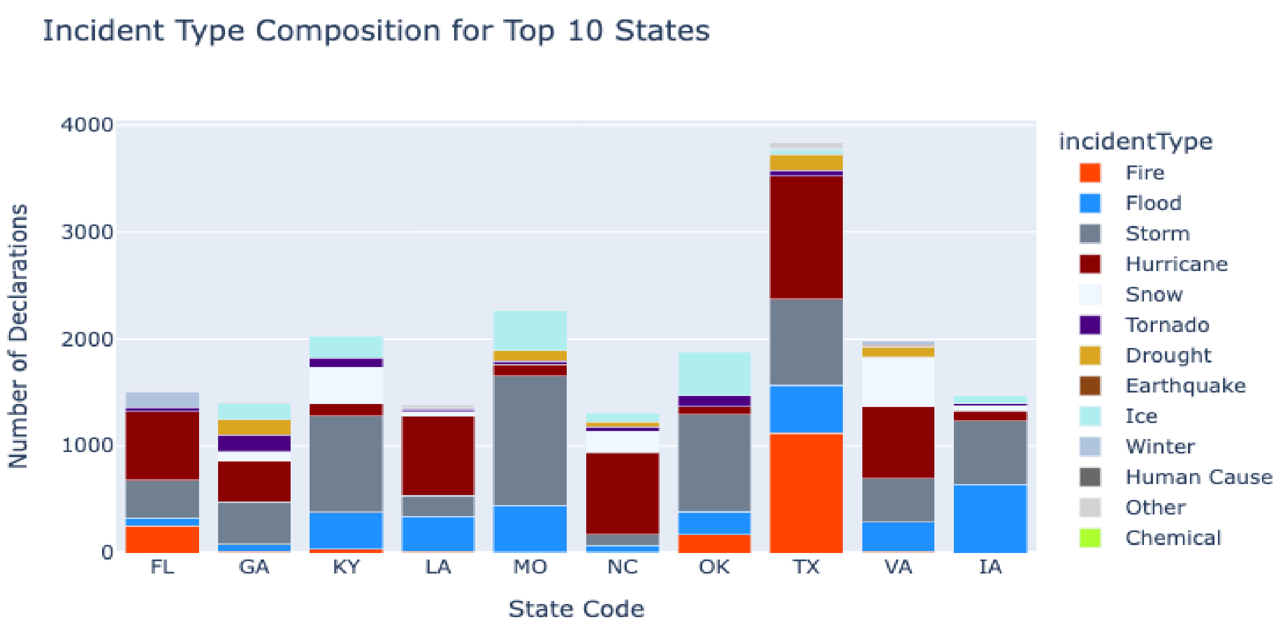
- **The Fire Belt (West):** California (CA) shows a clear statistical dominance in **Fire** declarations. This is a result of prolonged drought periods combined with high-density wildland-urban interfaces.
- **The Hurricane Alley (Southeast):** Florida (FL), Louisiana (LA), and North Carolina (NC) exhibit the highest density of **Hurricane** and **Coastal Storm** declarations.
- **The Multi-Threat Profile (Texas):** Texas (TX) is a unique case study in this analysis. It consistently ranks in the top tier for **Storms, Floods, Fire, and Hurricanes**, making it the most resource-intensive state in terms of federal disaster management.

5.3 Density Visualization (Heatmap)

The **Heatmap** serves as a "Quick-Look" matrix. While the national average for most disasters is low, the "Heat" (darker colors) concentrated in specific cells proves that federal disaster policy must remain highly regionalized to be effective.

Section 5.4 (State vs. Incident):





6. Disaster Assistance Program Analysis

Goal: To analyze the relationship between the nature of a disaster and the type of federal financial aid activated.

6.1 Programmatic Definitions

The two primary pillars of federal aid analyzed in this milestone are:

- **Individual Housing (IH):** Targeted toward private residents to help with temporary housing and home repairs.
- **Public Assistance (PA):** Targeted toward state and local governments to repair roads, bridges, and public utilities.

6.2 The "Assistance Gap"

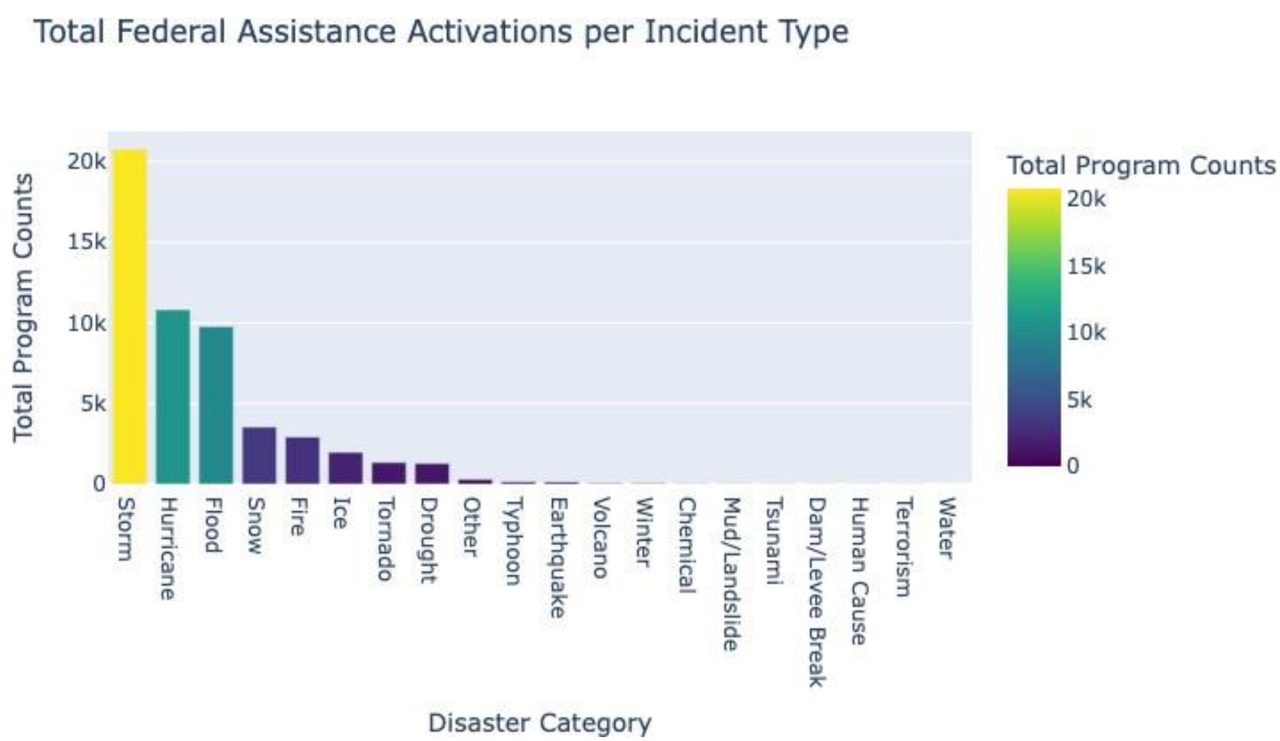
The **Grouped Bar Chart** reveals a significant trend: **Public Assistance (PA)** is triggered significantly more often than **Individual Assistance (IH)** across almost every category.

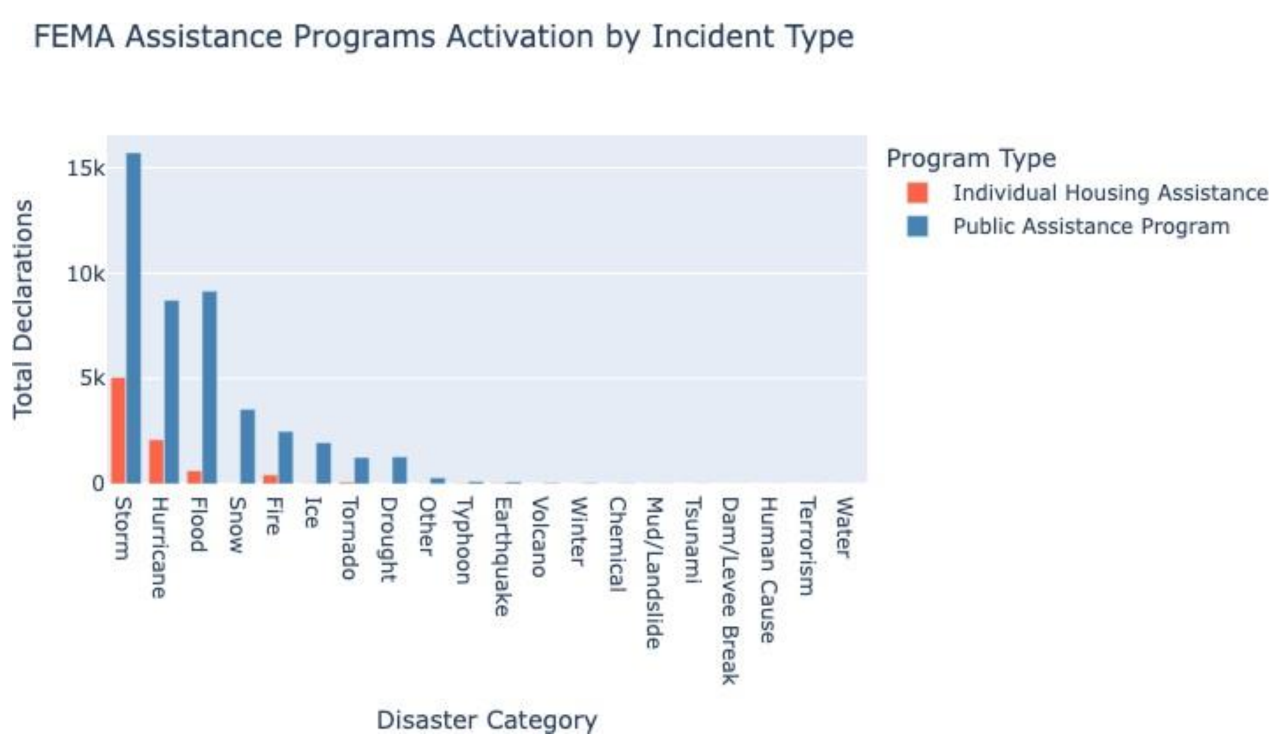
- **Infrastructural Burden:** For incidents like "Snow" or "Ice," PA is often the only program activated, as the primary federal role is helping cities clear roads rather than assisting individuals with home damage.
- **High-Intensity Exceptions:** Only in high-impact events like **Hurricanes** and **Floods** does the activation of Individual Housing aid approach the levels of Public Assistance.

6.3 Conclusion of Analysis

The correlation analysis proves that the federal government's primary role in disaster recovery is the restoration of public infrastructure. This insight is vital for policymakers to understand where the greatest financial burden lies during the recovery phase of different disaster types.

Section 6.4 (Assistance Programs):





7. Technical Methodology

This section outlines the logic and libraries used to transform raw data into the analytical insights presented in the previous sections.

7.1 Data Manipulation with Pandas

The core of the analysis relied on the **Pandas** library for complex data restructuring:

- **Pivot Tables:** Used to create the $State \times Incident$ matrix required for the heatmap.
- **Melting Data:** To create the **Grouped Bar Chart**, the wide-format assistance data was "melted" into a long-format. This allowed the visualization library to treat "Program Type" as a categorical variable for side-by-side comparison.
- **Logical Mapping:** Applied vectorized operations to convert categorical "Yes/No" strings into binary integers ($1/0$), enabling the summation of assistance activations.

7.2 Advanced Visualization Logic

- **Plotly Express:** Chosen for its ability to handle multi-dimensional data. For example, in the **Stacked Bar Chart**, Plotly allowed for the simultaneous display of state-level totals and incident-specific contributions through color mapping.
- **Seaborn Heatmaps:** Utilized for density visualization. By applying a sequential color palette (e.g., YlOrRd), we were able to highlight "hot" cells where specific disasters significantly deviated from the national average.

8. Conclusion

The completion of Milestone 4 marks the culmination of the US Natural Disasters analysis project (1953–2017). By synthesizing the findings from all four phases, we have developed a comprehensive view of the American disaster landscape.

8.1 Summary of Findings

- **Chronological Risk:** As established in Milestone 2, disaster frequency is not static; it is increasing, with specific months like August and September posing the highest risk.
- **Geographic Vulnerability:** Milestone 3 and 4 confirmed that geography dictates destiny. Coastal states are bound to hydrological threats, while the West is increasingly defined by Fire incidents.
- **Federal Strategy:** The data indicates that the federal response is structurally geared toward **Public Assistance (PA)**. While individual aid is critical for recovery, the vast majority of declared disasters focus on the restoration of community infrastructure.

Access the Live Dashboard here: https://u-s-natural-disaster-dashboard.vercel.app